Deteção de Fraude

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Definição do Problema

- Problema:

  Utilizar um dataset de transações financeiras no ecossistema mobile simuladas

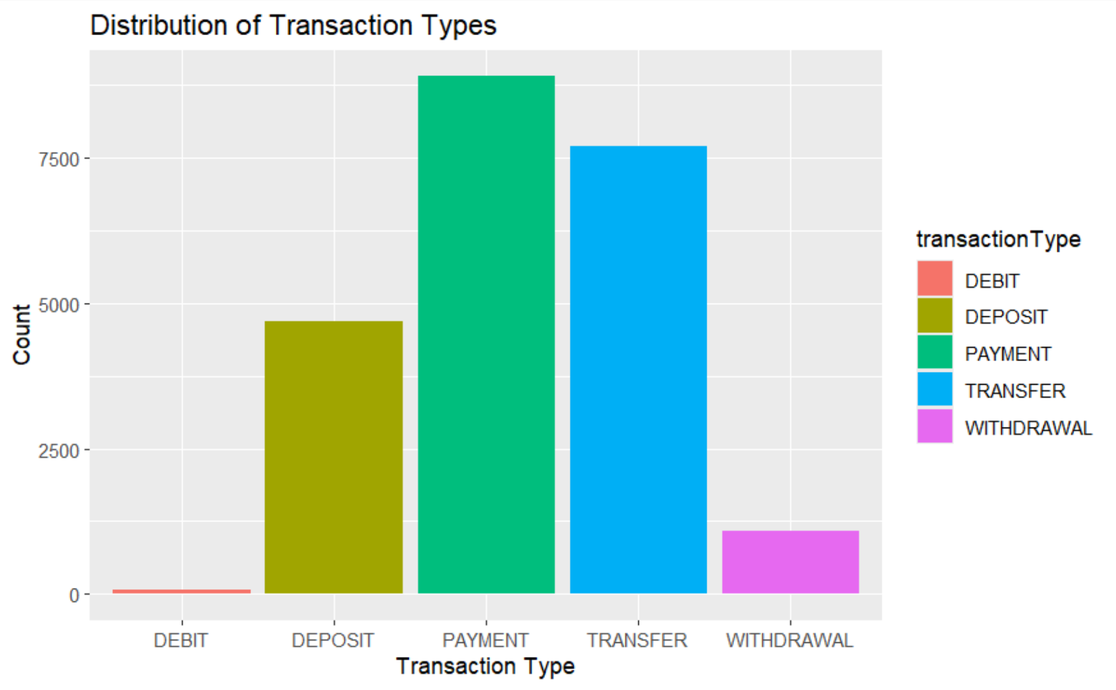
- Proposta:

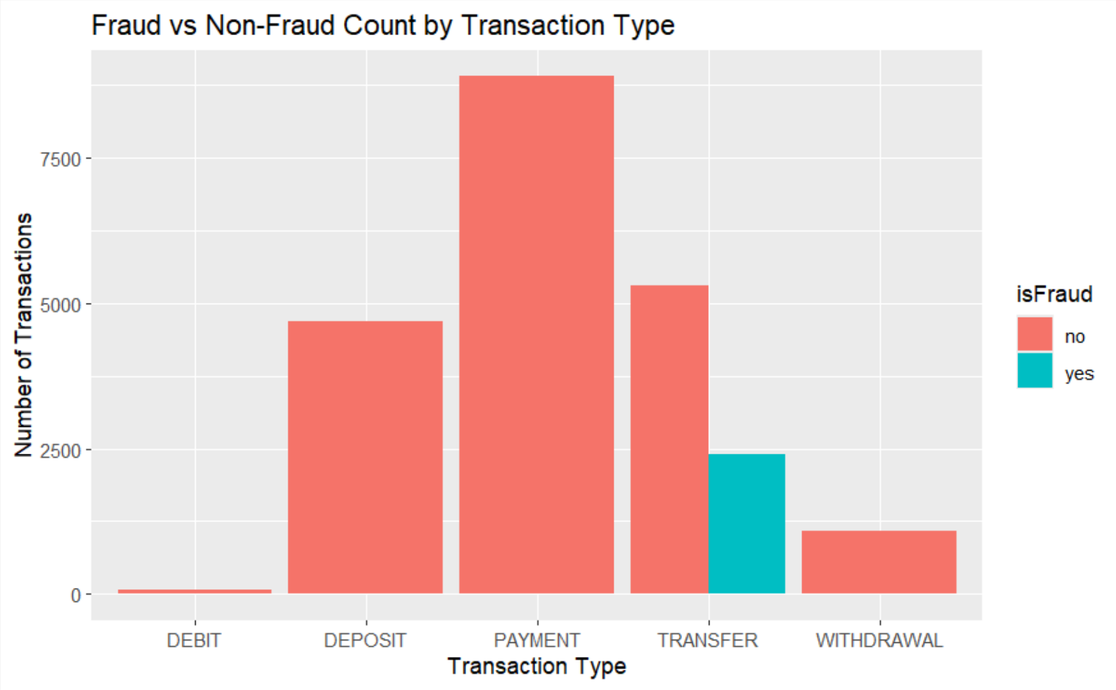
  Criar um modelo de machine learning capaz de detetar se uma transação é fraudulenta ou não

Compreensão dos Dados

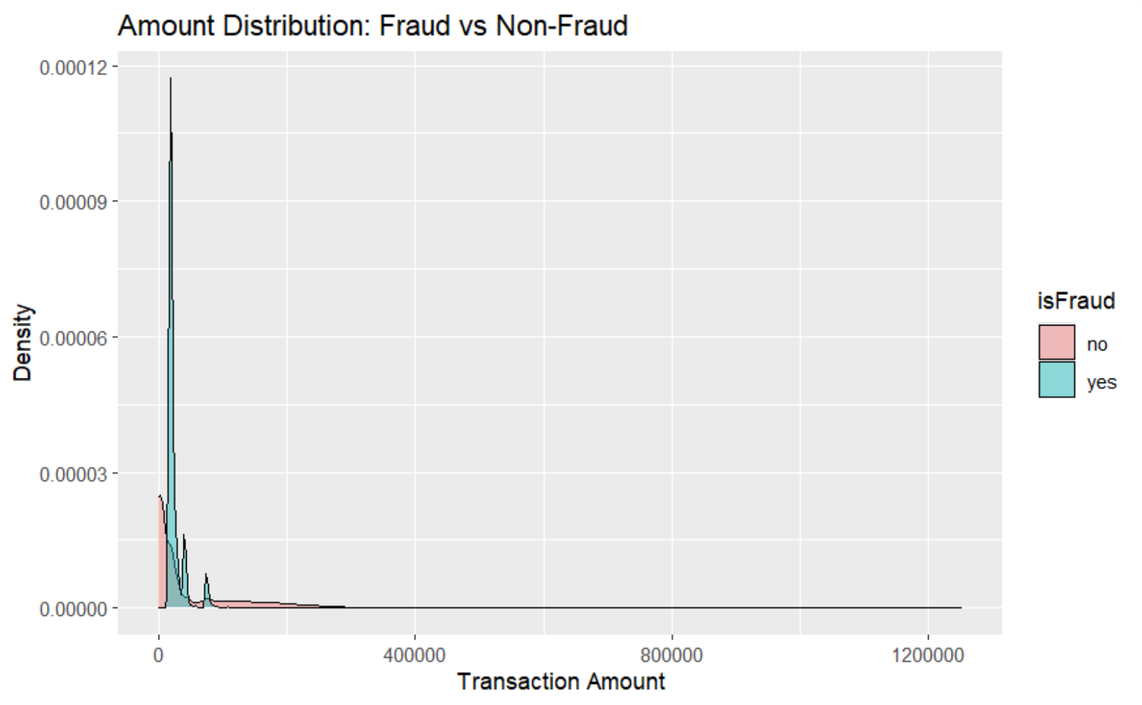
Attribute	Description	Data Type	Scale
transactionID	Unique identifier for the transaction	Categorical	Nominal
step	Time step; 1 step = 1 hour	Numeric	Ratio
transactionType	Transaction category (deposit, withdrawal, transfer, payment, debit)	Categorical	Nominal
amount	Monetary amount of the transaction	Numeric	Ratio
initiator	ID of the user initiating the transaction	Categorical	Nominal
oldBalInitiator	Initiator account balance before the transaction	Numeric	Ratio
newBalInitiator	Initiator account balance after the transaction	Numeric	Ratio
recipient	ID of the user receiving the transaction	Categorical	Nominal
oldBalRecipient	Recipient account balance before the transaction	Numeric	Ratio
newBalRecipient	Recipient account balance after the transaction	Numeric	Ratio
isFraud	Fraud label (1 = fraud, 0 = legitimate)	Categorical	Nominal

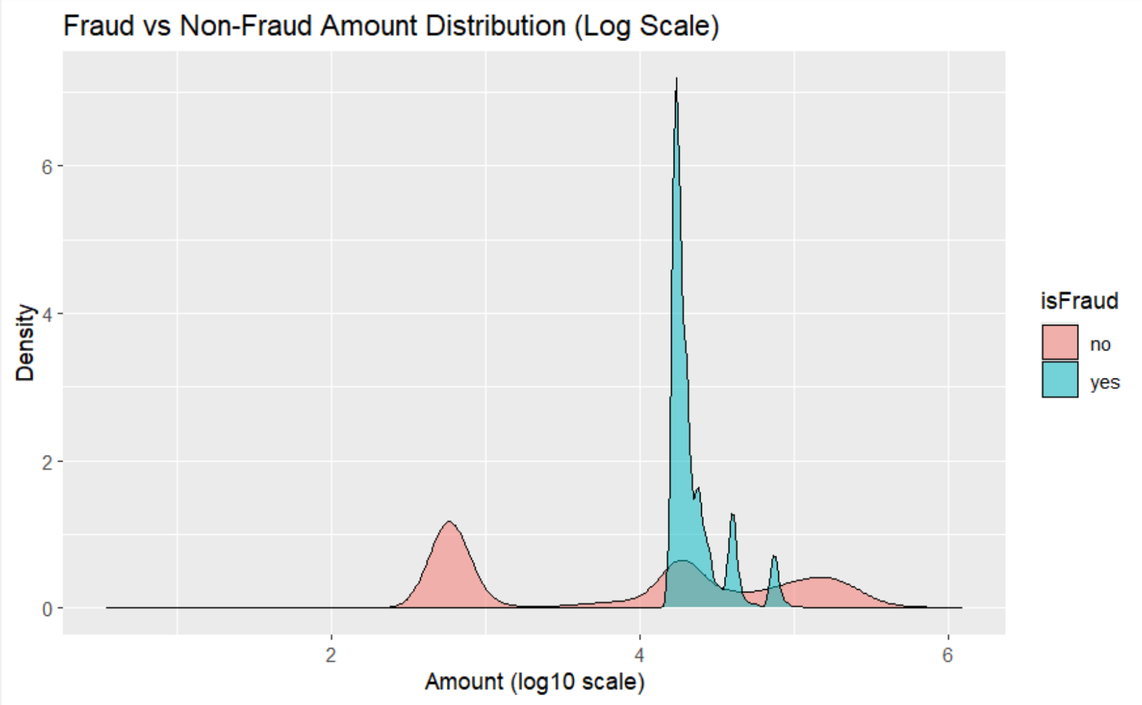
Análise do tipo de transações



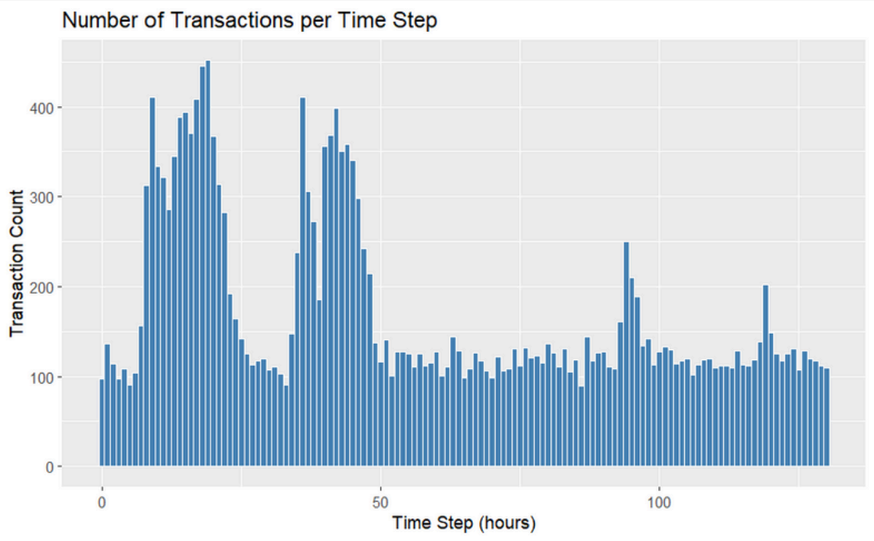


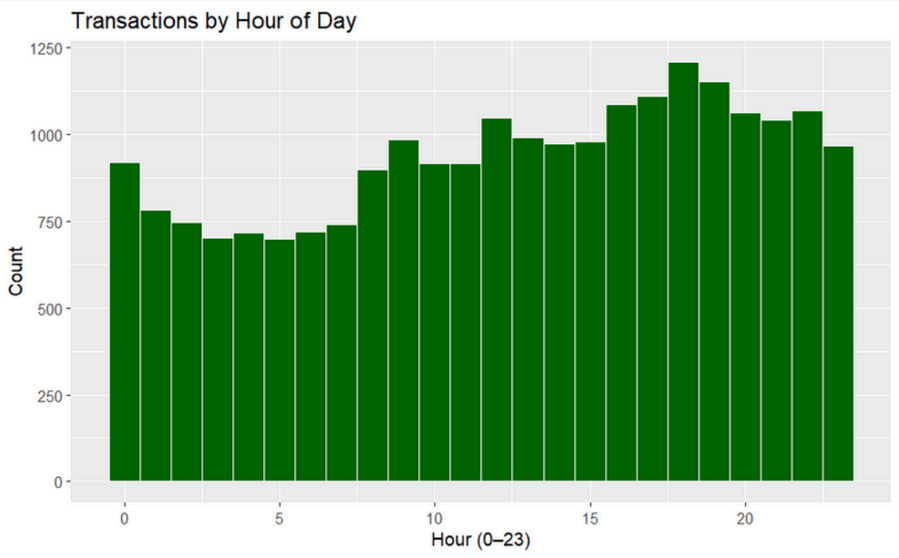
Análise do valor da transação

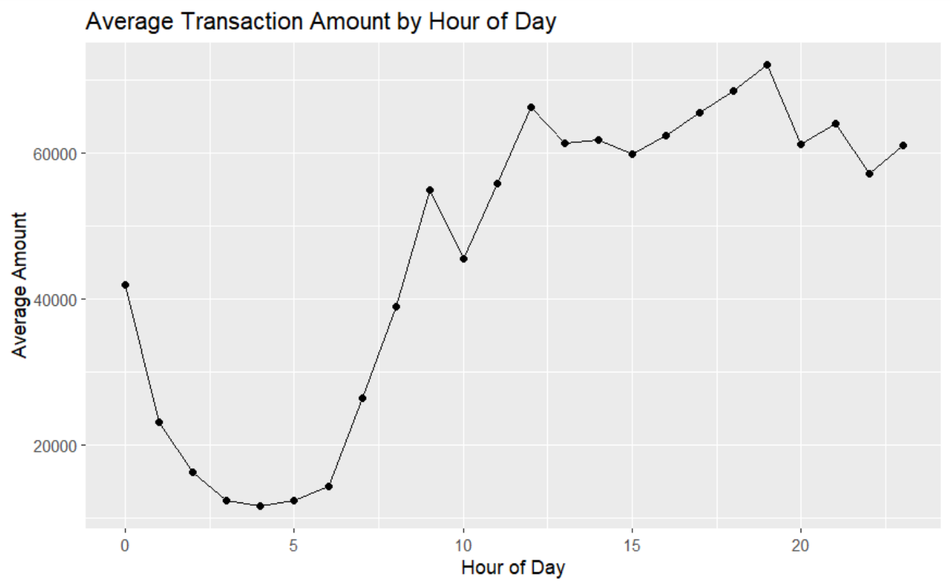


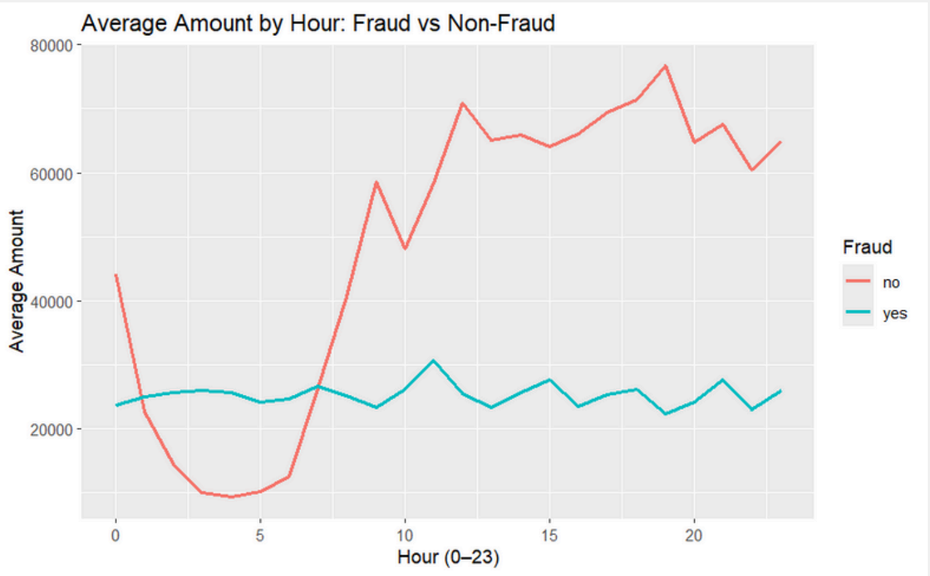


Análise Temporal

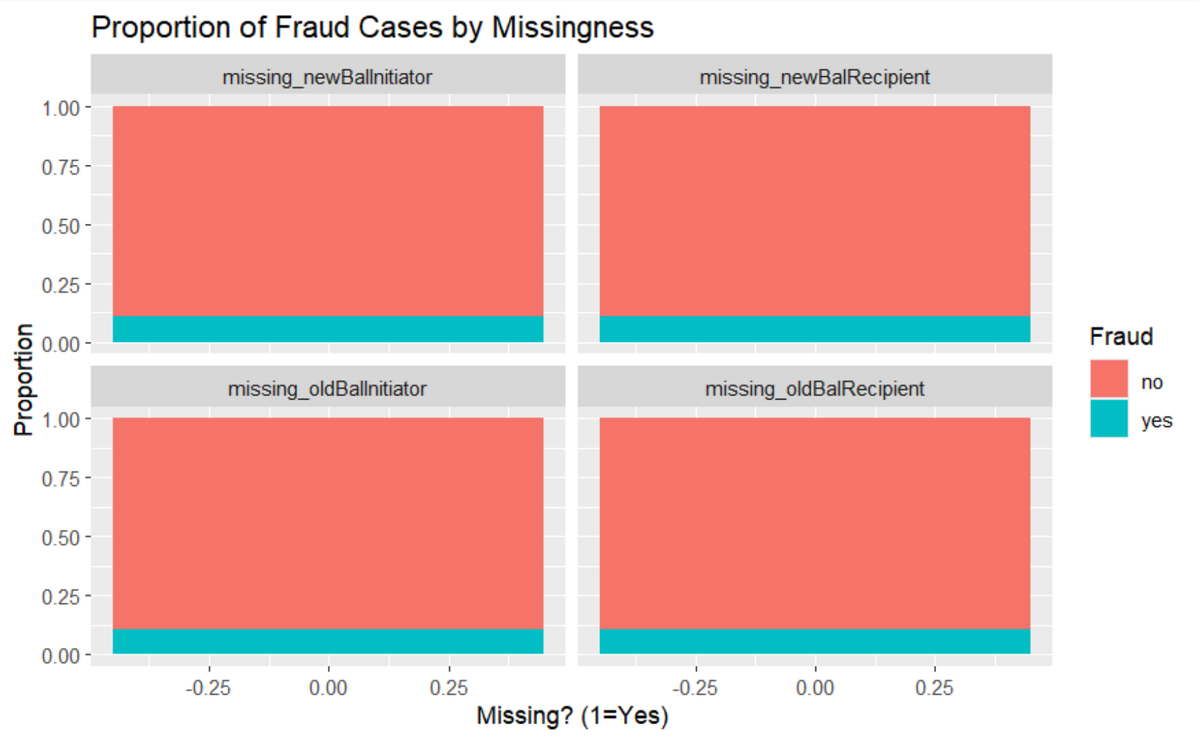


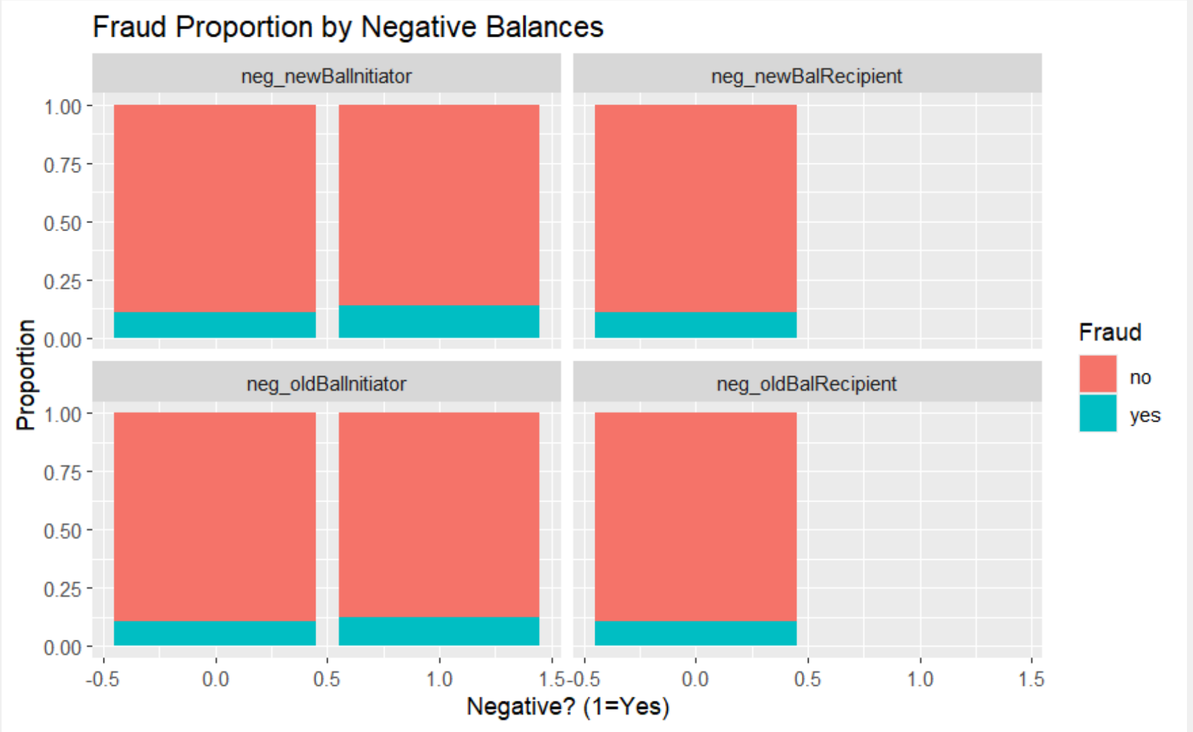






Análise dos casos especiais



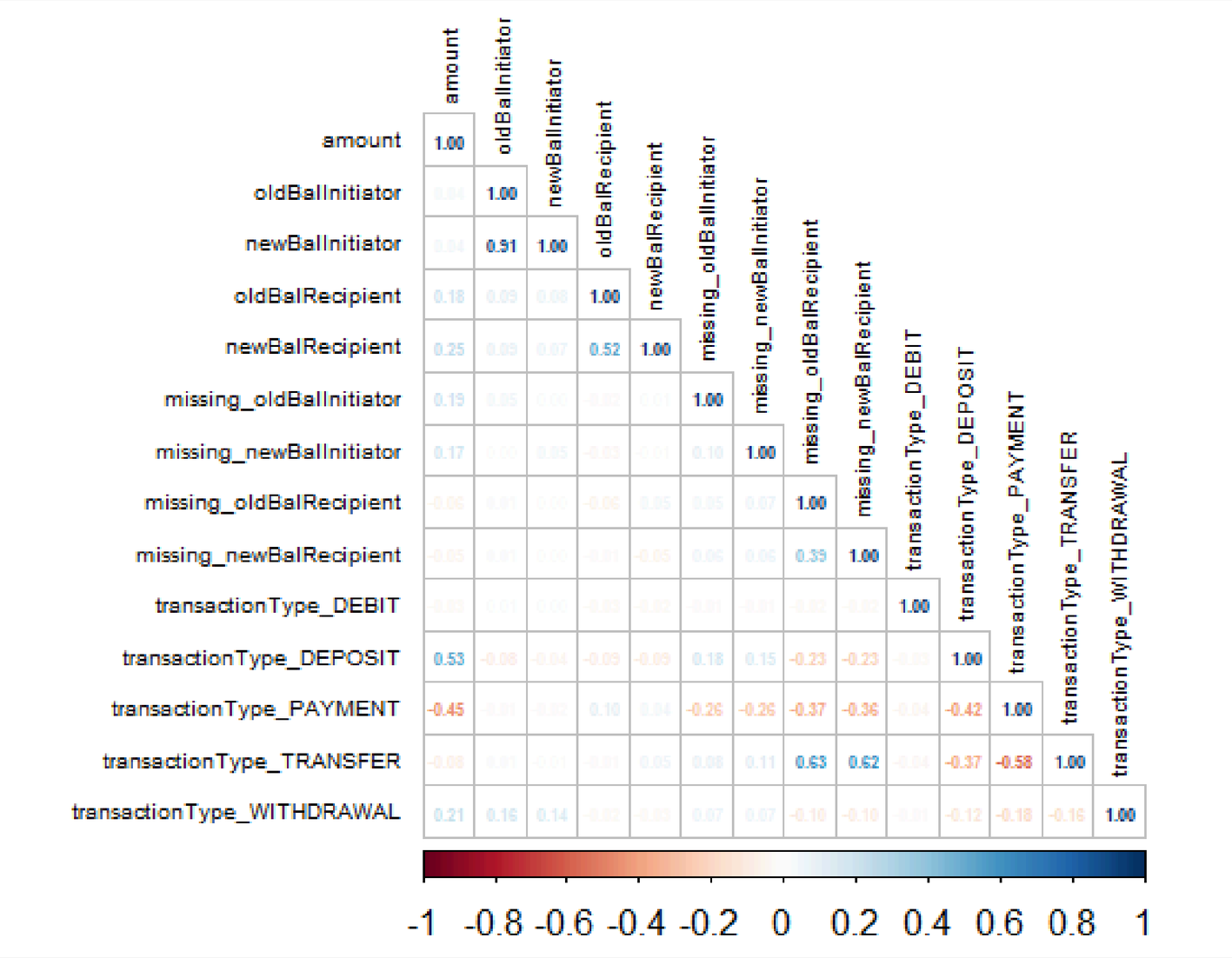


Preparação dos Dados

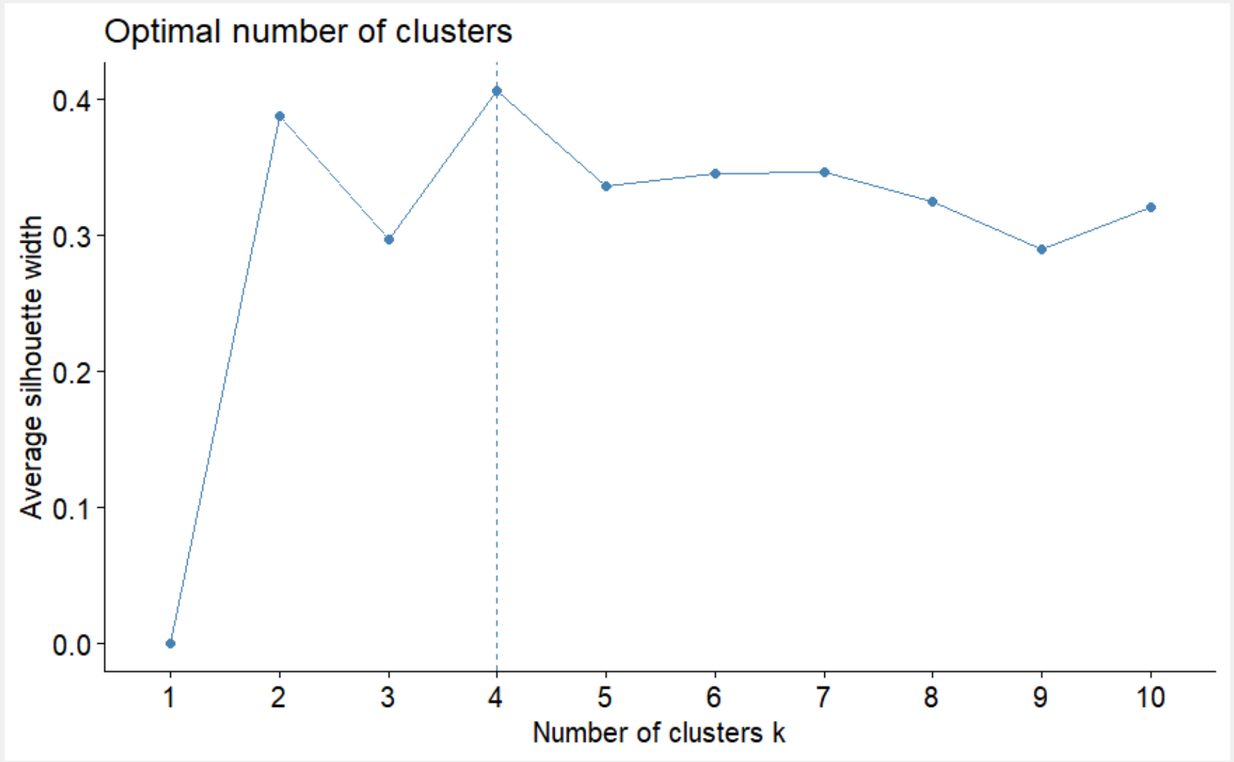
```
'data.frame':   22440 obs. of  11 variables:
 $ transactionID  : int  122235 122236 122237 122238 122239 122240 122241 122242 122243 122244 ...
 $ step           : int  0 0 0 0 0 0 0 0 0 0 ...
 $ transactionType: chr  "PAYMENT" "PAYMENT" "PAYMENT" "PAYMENT" ...
 $ amount         : num  538 602 598 573 509 ...
 $ initiator      : num  4.13e+15 4.70e+15 4.17e+15 4.80e+15 4.26e+15 ...
 $ oldBalInitiator: chr  "22.47" "1202.7" "173370.94" "120941.28" ...
 $ newBalInitiator: chr  "22.47" "600.45" "172773.03" "120368.22" ...
 $ recipient      : chr  "60-0001777" "00-0003843" "86-0001469" "66-0008129" ...
 $ oldBalRecipient: chr  "0" "0" "0" "0" ...
 $ newBalRecipient: chr  "0" "602.25" "597.9" "573.06" ...
 $ isFraud        : chr  "no" "no" "no" "no" ...
```

```
tibble [17,951 × 15] (S3: tbl_df/tbl/data.frame)
 $ amount                    : num [1:17951] -0.572 -0.571 -0.571 -0.572 -0.572 ...
 $ oldBalInitiator           : num [1:17951] -1.92 -1.92 -1.83 -1.91 -1.92 ...
 $ newBalInitiator           : num [1:17951] -1.96 -1.96 -1.86 -1.95 -1.96 ...
 $ oldBalRecipient           : num [1:17951] -0.47 -0.47 -0.47 -0.47 -0.466 ...
 $ newBalRecipient           : num [1:17951] -0.408 -0.406 -0.406 -0.406 -0.406 ...
 $ missing_oldBalInitiator   : num [1:17951] 0 0 0 0 0 0 0 0 0 0 ...
 $ missing_newBalInitiator   : num [1:17951] 0 0 0 0 0 0 0 0 0 0 ...
 $ missing_oldBalRecipient   : num [1:17951] 0 0 0 0 0 0 0 0 0 0 ...
 $ missing_newBalRecipient   : num [1:17951] 0 0 0 0 0 0 0 0 0 0 ...
 $ isFraud                   : Factor w/ 2 levels "yes","no": 2 2 2 2 2 2 2 2 2 2 ...
 $ transactionType_DEBIT     : num [1:17951] 0 0 0 0 0 0 0 0 0 0 ...
 $ transactionType_DEPOSIT   : num [1:17951] 0 0 0 0 0 0 0 0 0 0 ...
 $ transactionType_PAYMENT   : num [1:17951] 1 1 1 1 1 1 1 1 1 1 ...
 $ transactionType_TRANSFER  : num [1:17951] 0 0 0 0 0 0 0 0 0 0 ...
 $ transactionType_WITHDRAWAL: num [1:17951] 0 0 0 0 0 0 0 0 0 0 ...
```

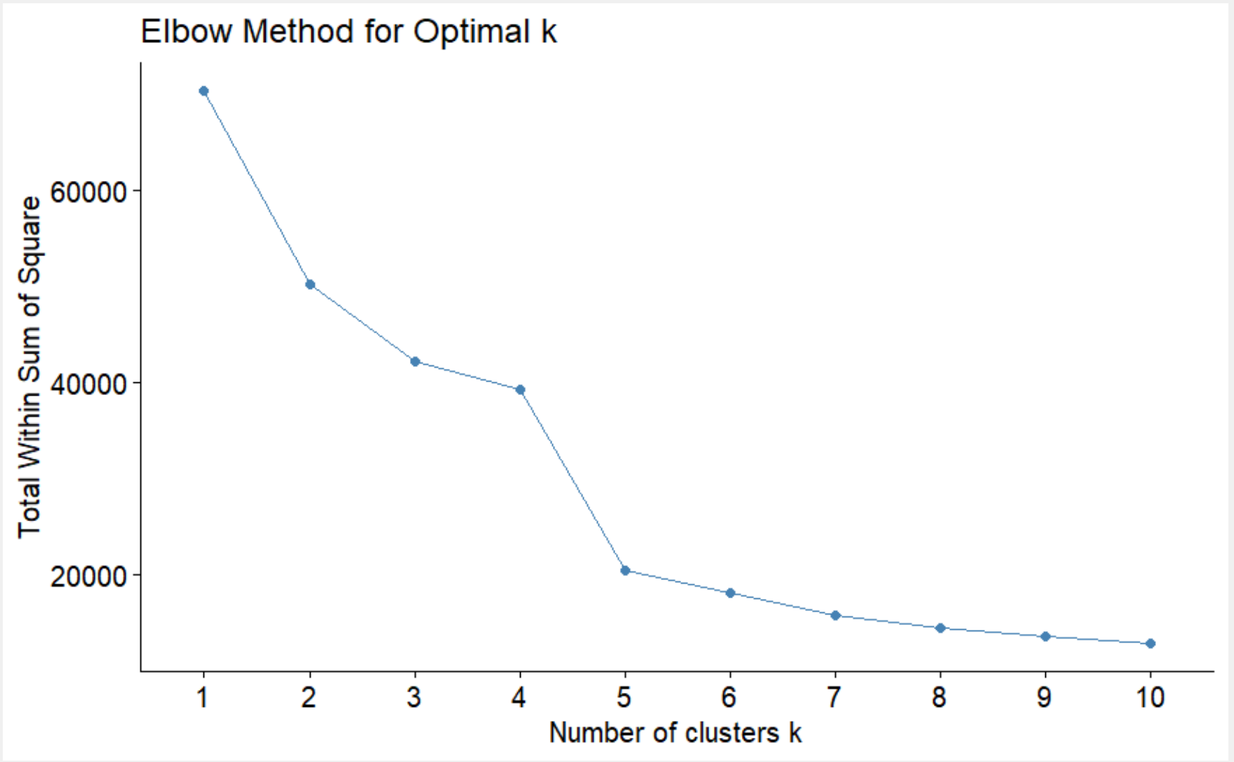
Correlação


amount
oldBalInitiator
newBalInitiator
oldBalRecipient
newBalRecipient
missing_oldBalInitiator
missing_newBalInitiator
missing_oldBalRecipient
missing_newBalRecipient
transactionType_DEBIT
transactionType_DEPOSIT
transactionType_PAYMENT
transactionType_TRANSFER
transactionType_WITHDRAWAL
-1 -0.8 -0.6 -0.4 -0.2 0 0.2 0.4 0.6 0.8 1

Clustering

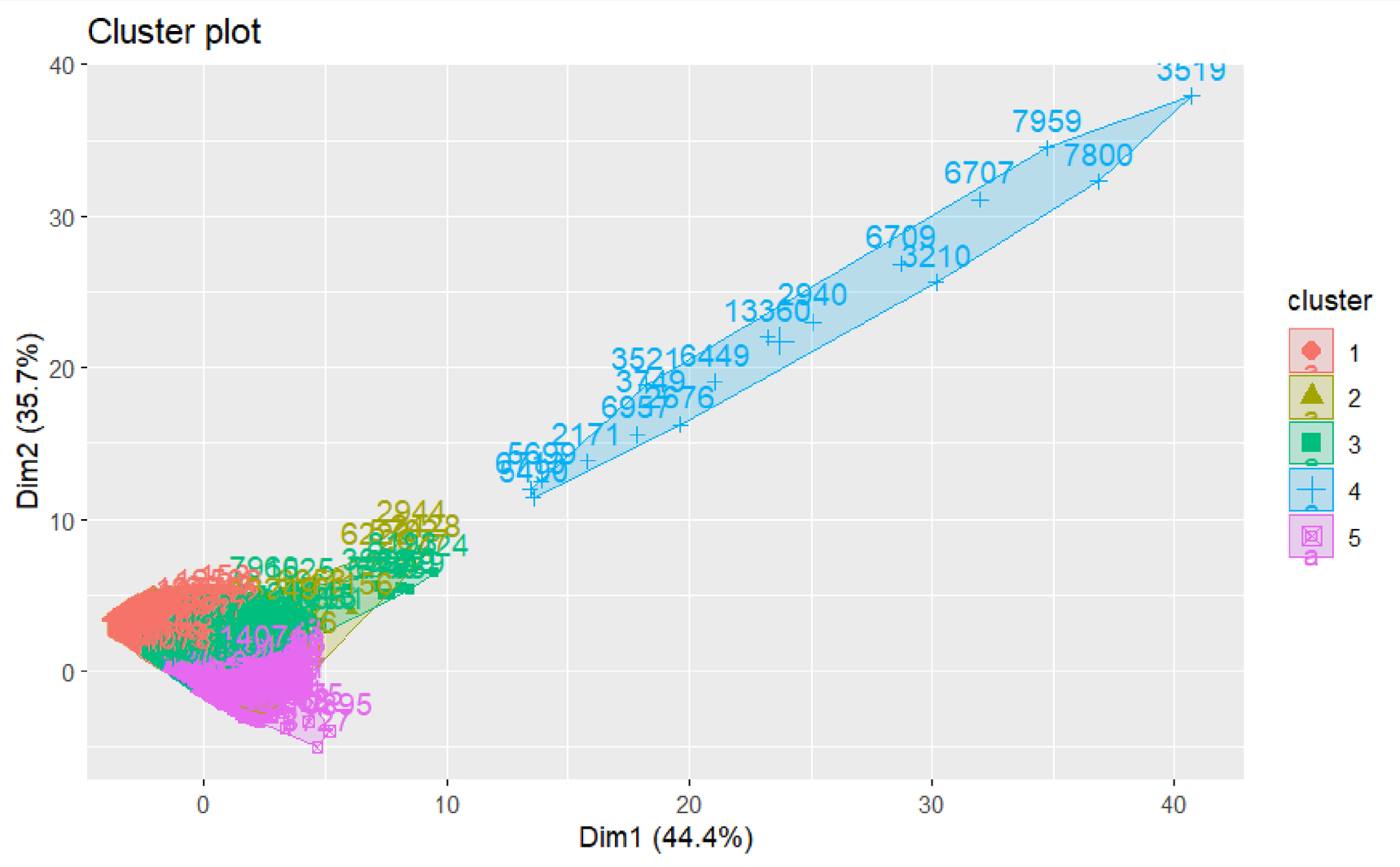


Silhouette

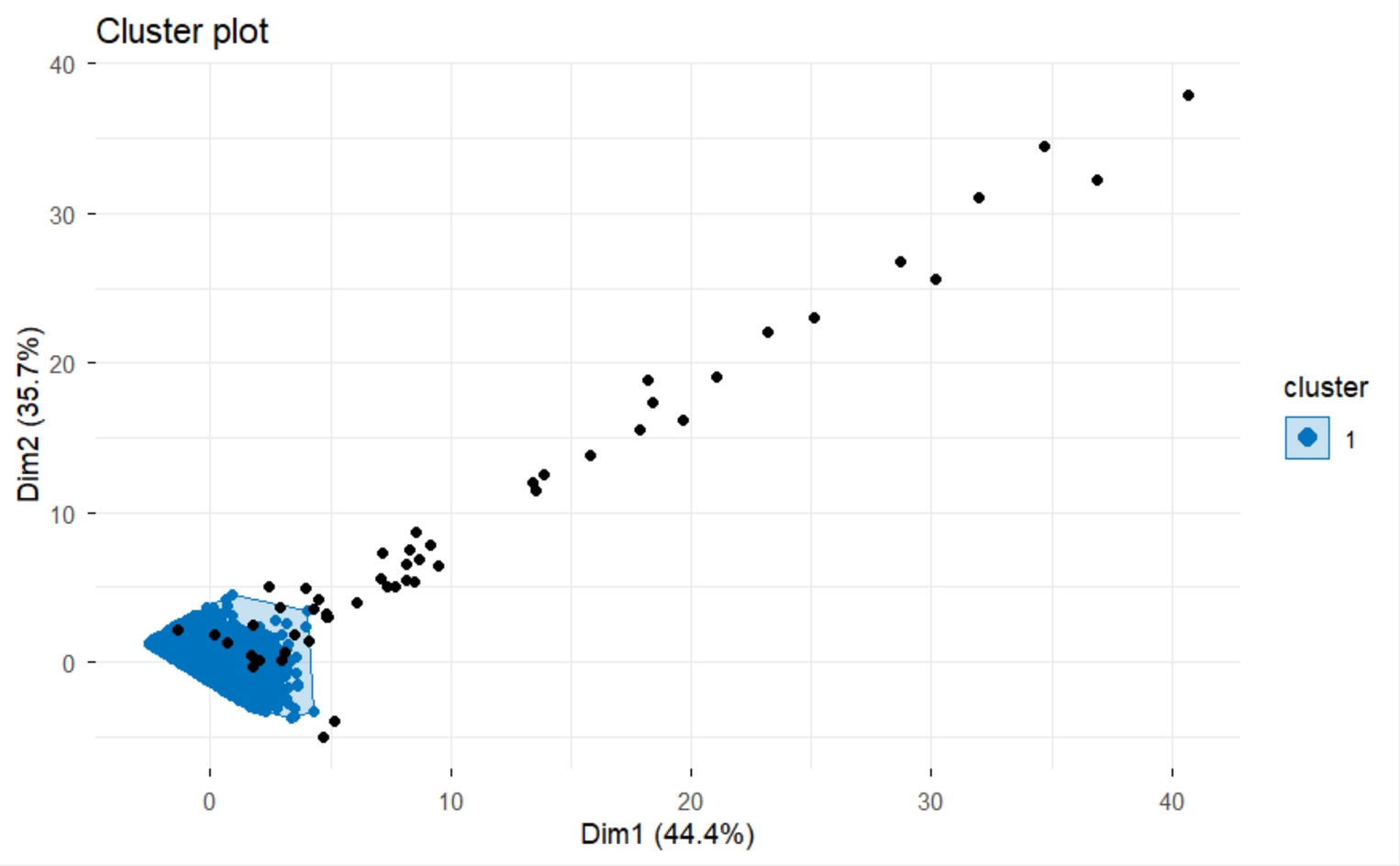


Elbow

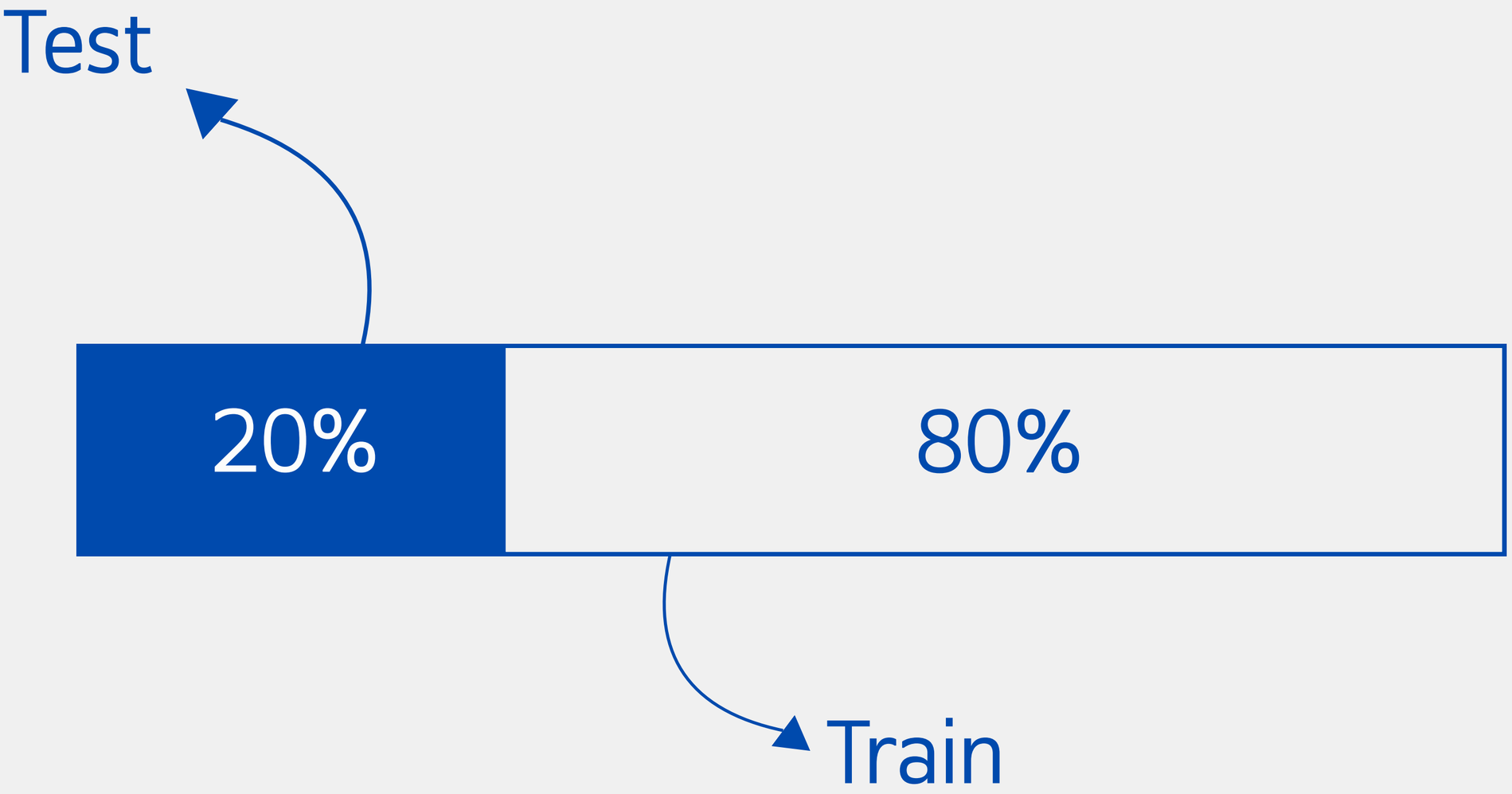
Clustering - K-means



Clustering - DBSCAN



Divisão do conjunto de dados



Modelação

1. SMOTE
2. Grid
3. Cross validation
4. Tunning - ROC AUC

SMOTE

isFraud <fctr>	n <int>	proportion <dbl>
yes	1925	0.1072364
no	16026	0.8927636

2 rows

isFraud <fctr>	n <int>	proportion <dbl>
yes	15400	0.4900401
no	16026	0.5099599

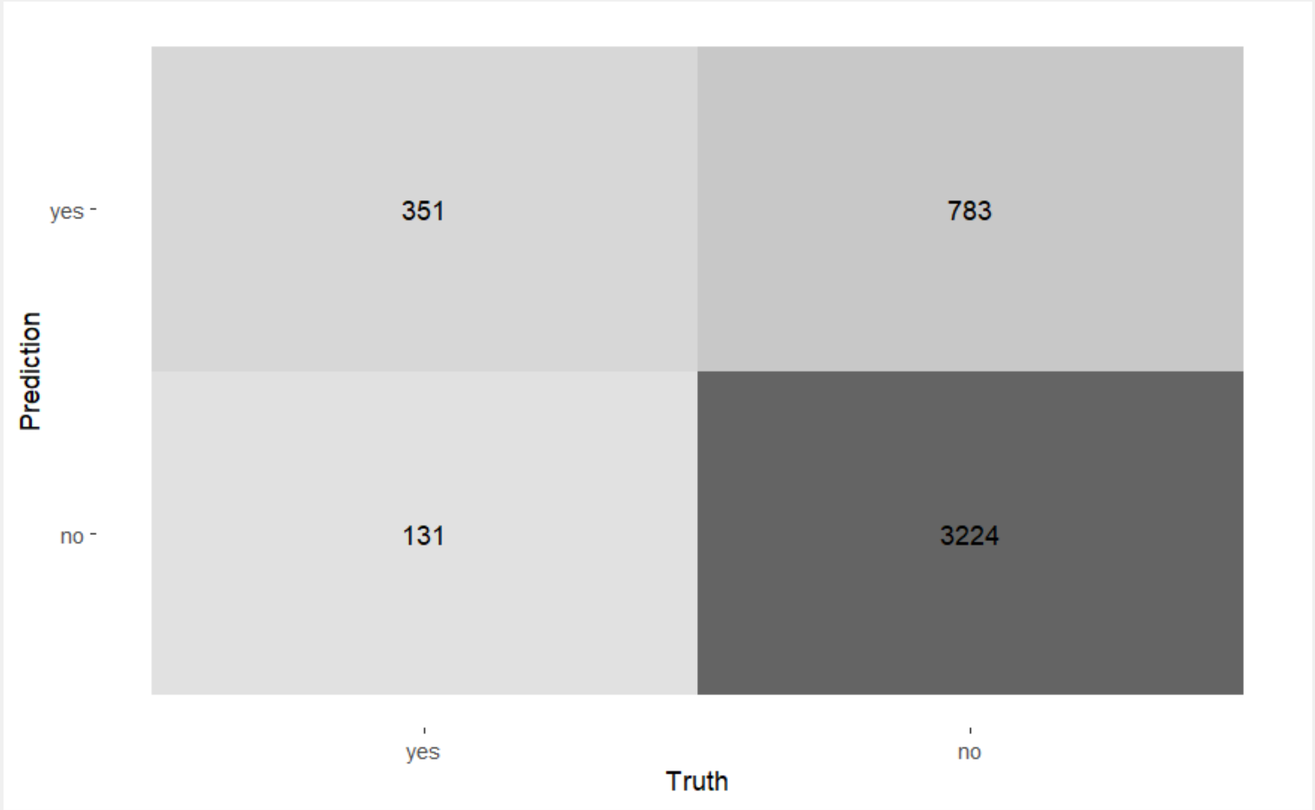
2 rows

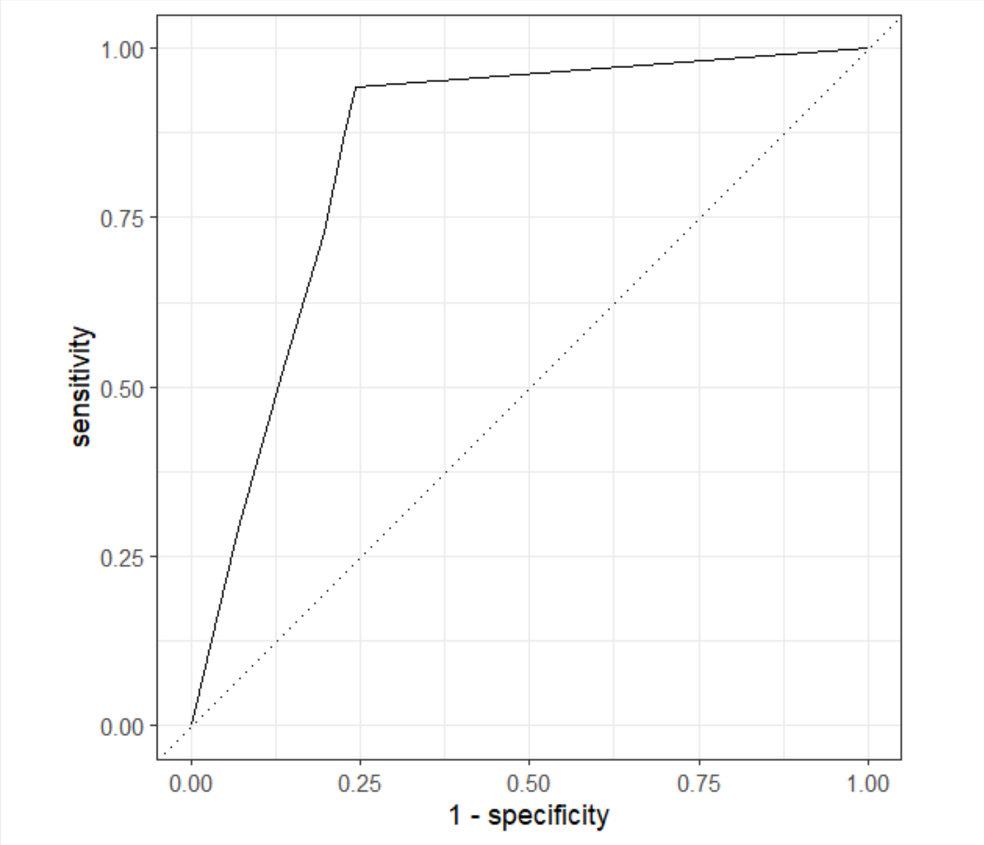
Modelos implementados

1. KNN
2. NB
3. Random Forest
4. SVM
5. Neural Network

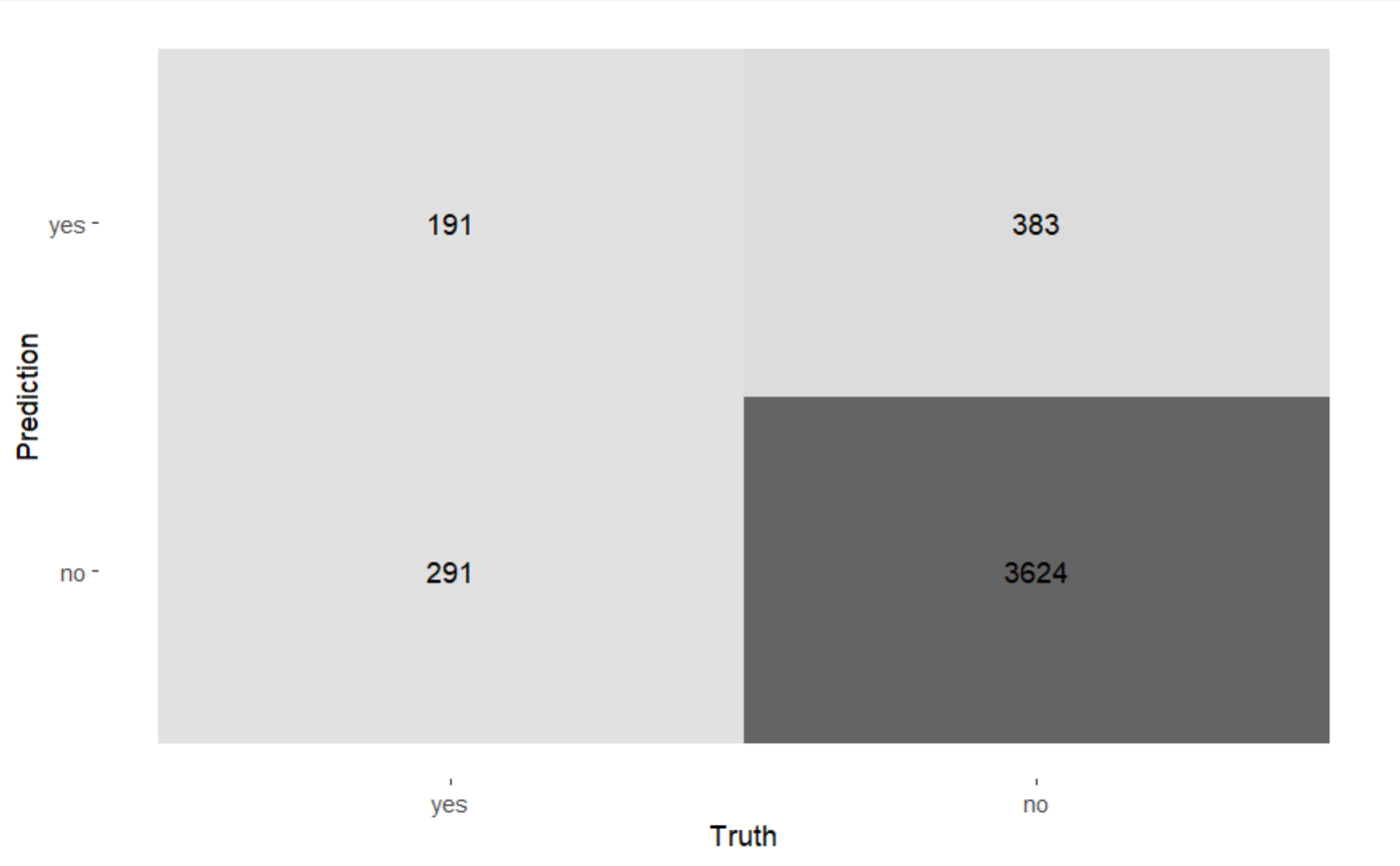
KNN

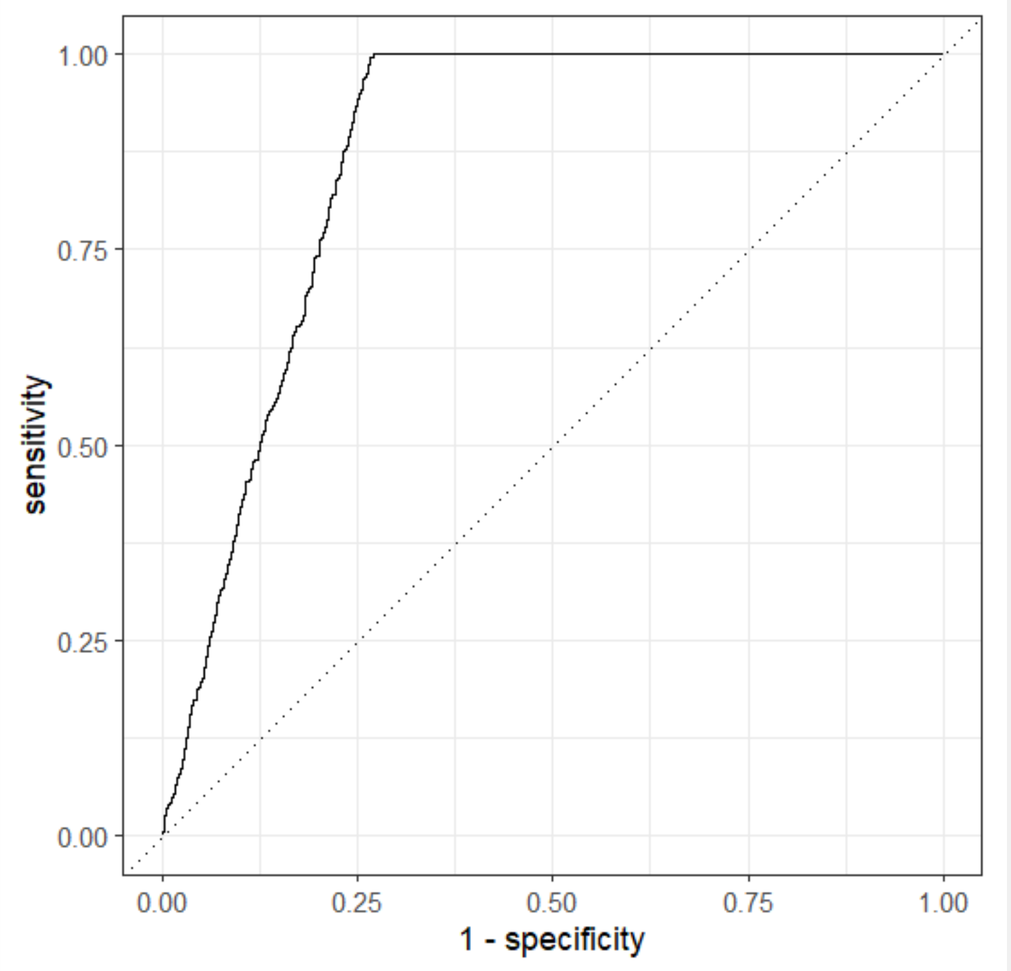
Parâmetro	Valores
Neighbours	3, 5, 7, 9, 11, 15, 21
weight_func	rectangular



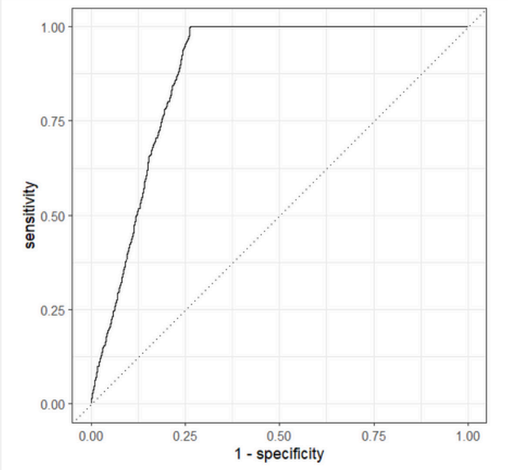


NB

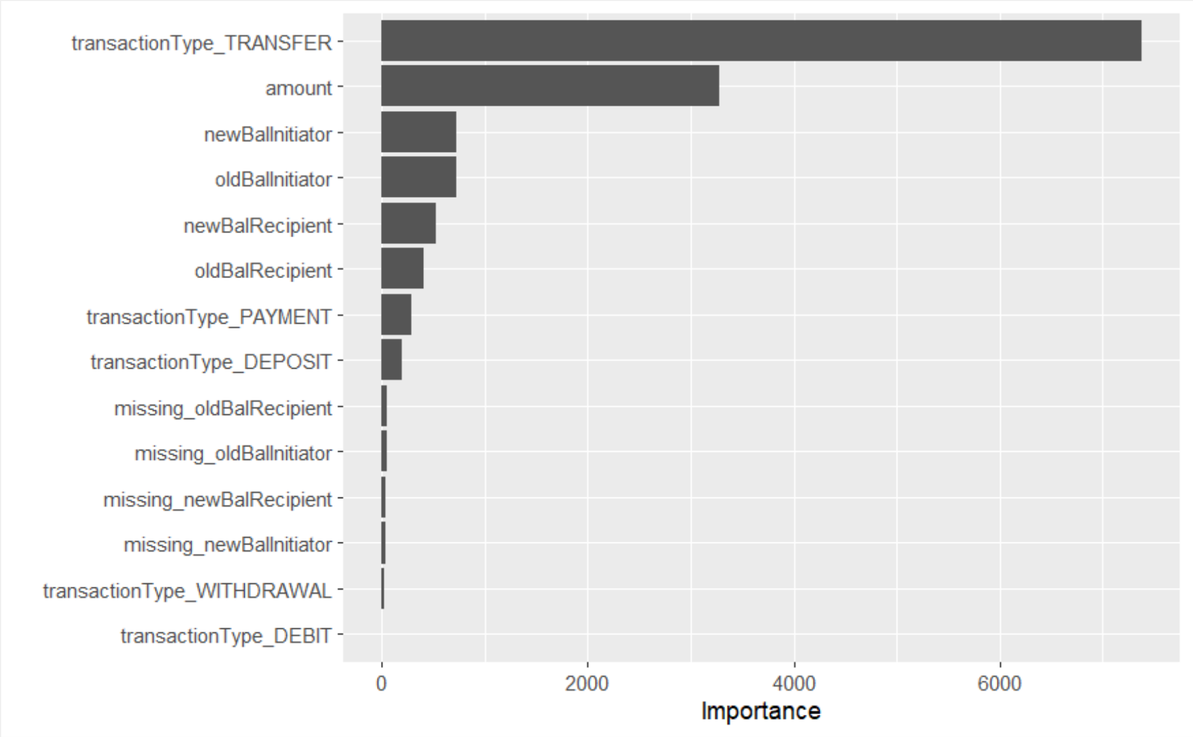

yes
no
Prediction
191
383
291
3624
yes
no
Truth

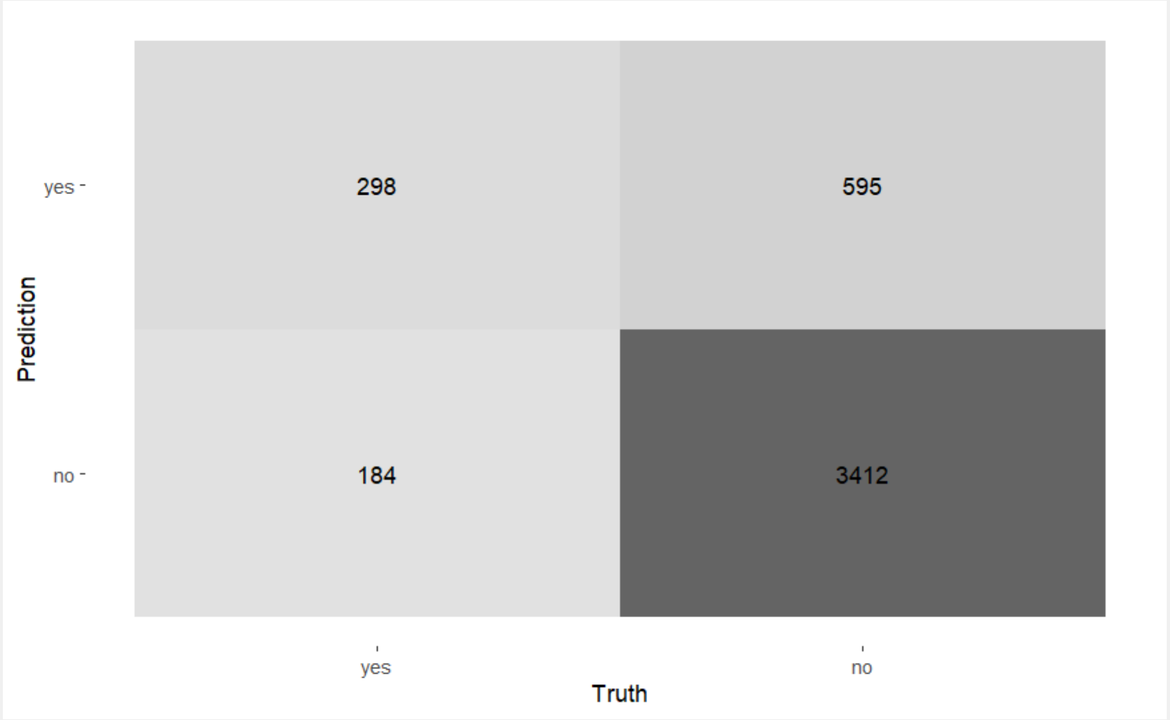

1.00
0.75
0.50
0.25
0.00
sensitivity
0.00
0.25
0.50
0.75
1.00
1 - specificity

Random Forest



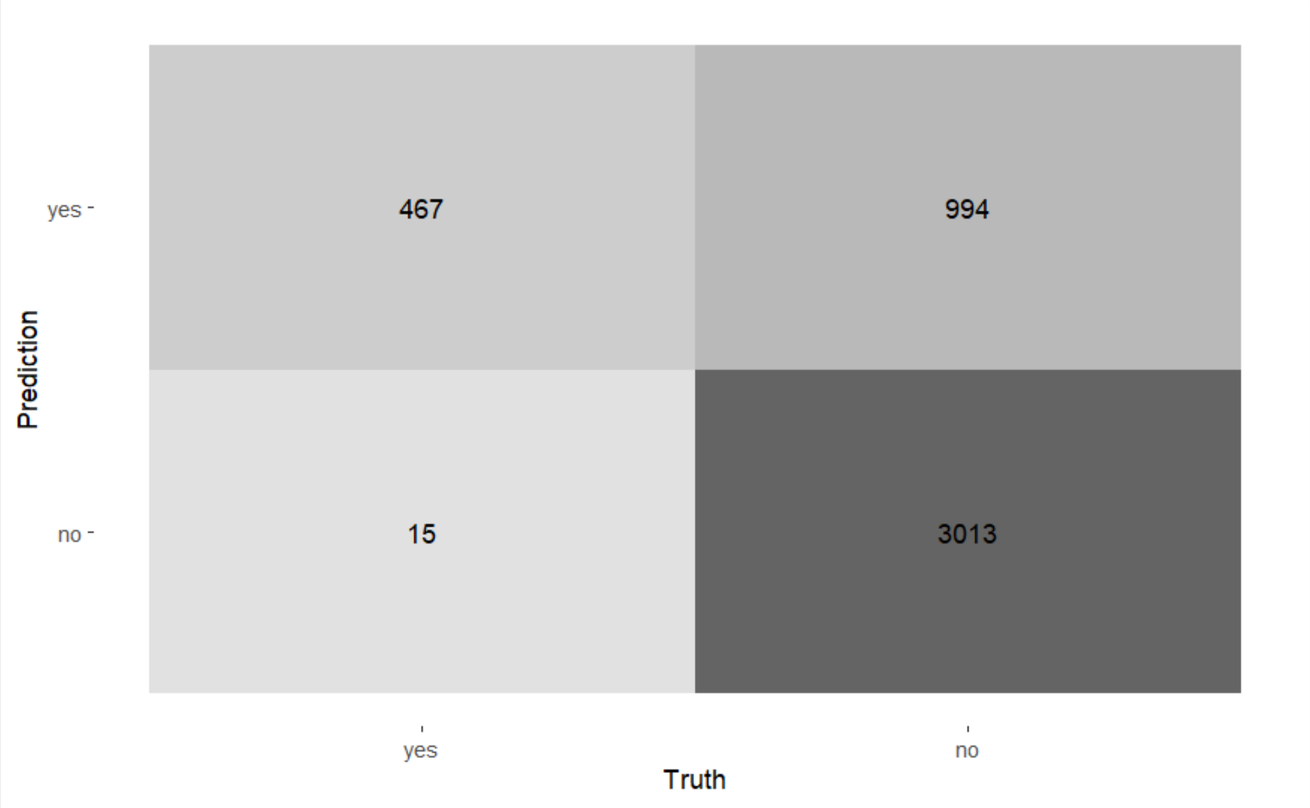
Parâmetro	Valores
mtry	3, 4, 5, 7, 10
min_n	2, 5, 10, 15, 20
trees	500

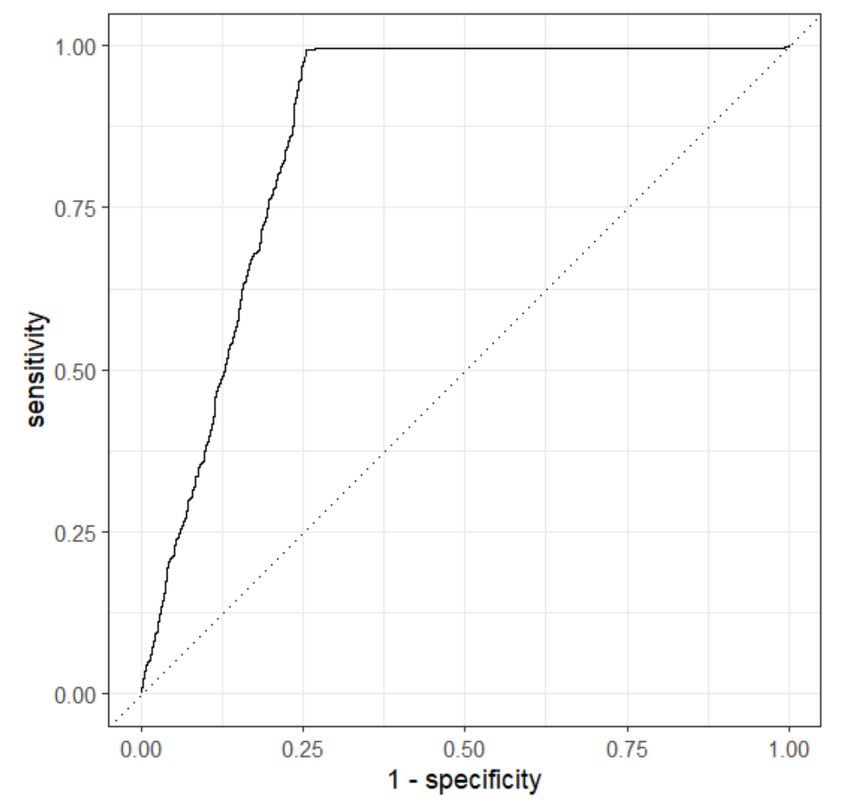




SVM

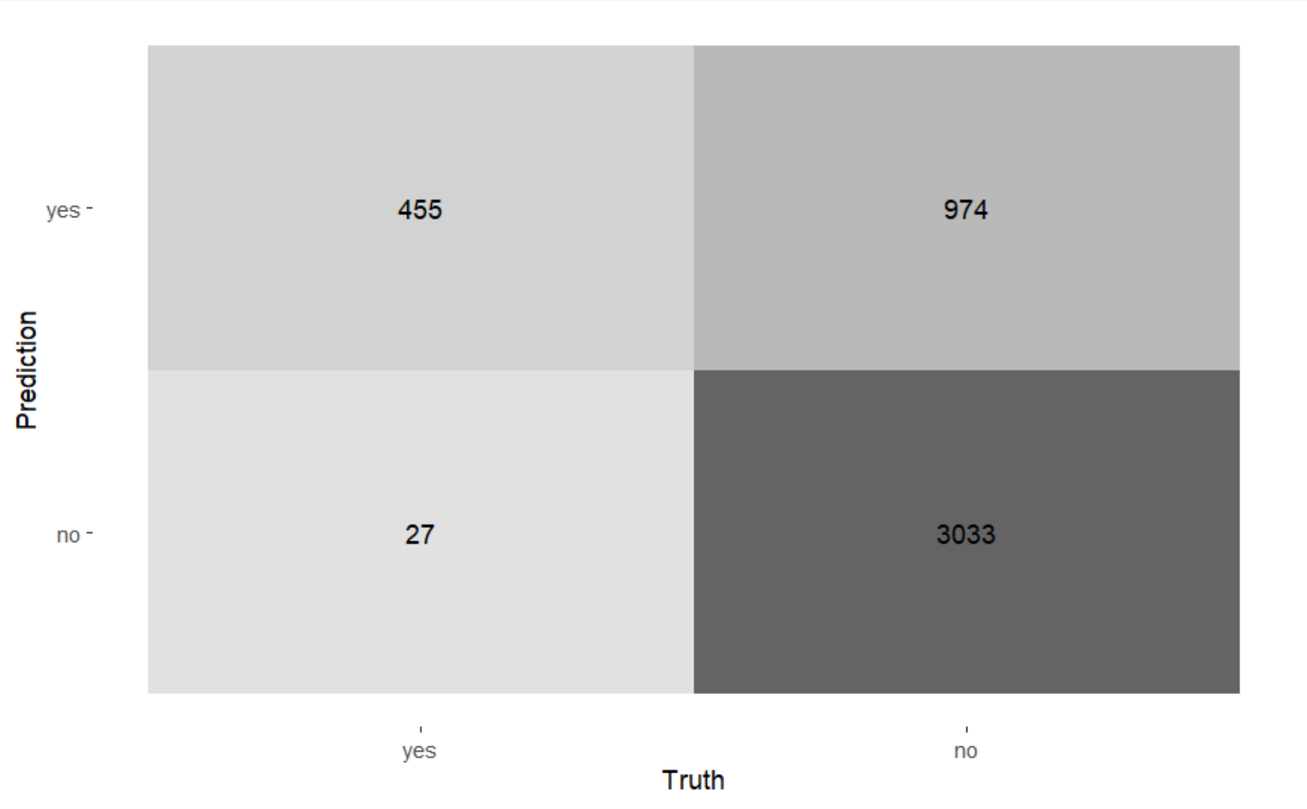
Parâmetro	Valores
cost	3, 4, 5, 7, 10
rbf_sigma	2, 5, 10, 15, 20

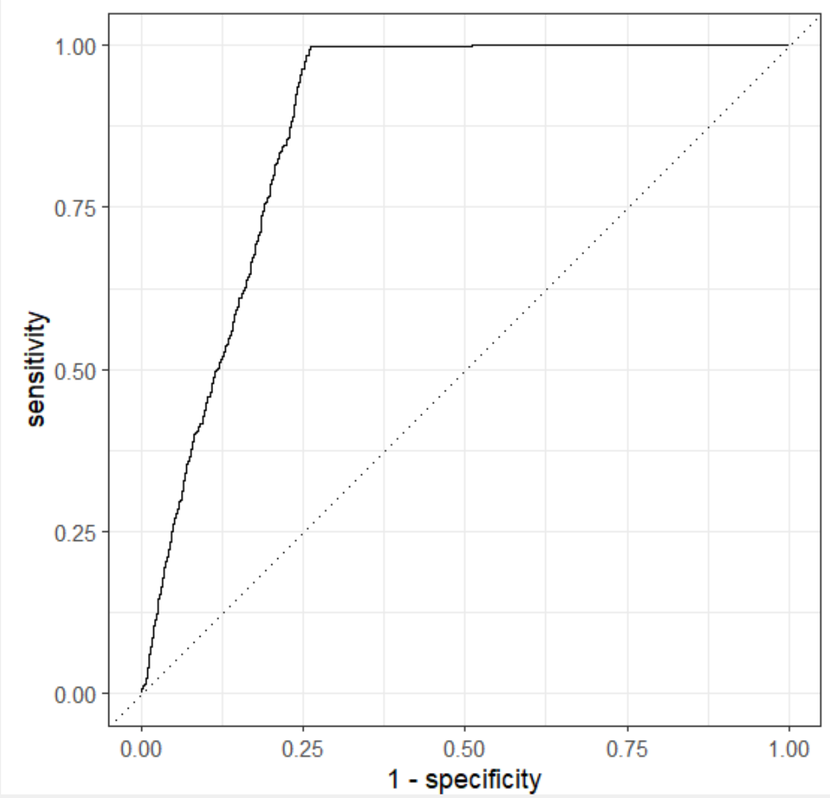




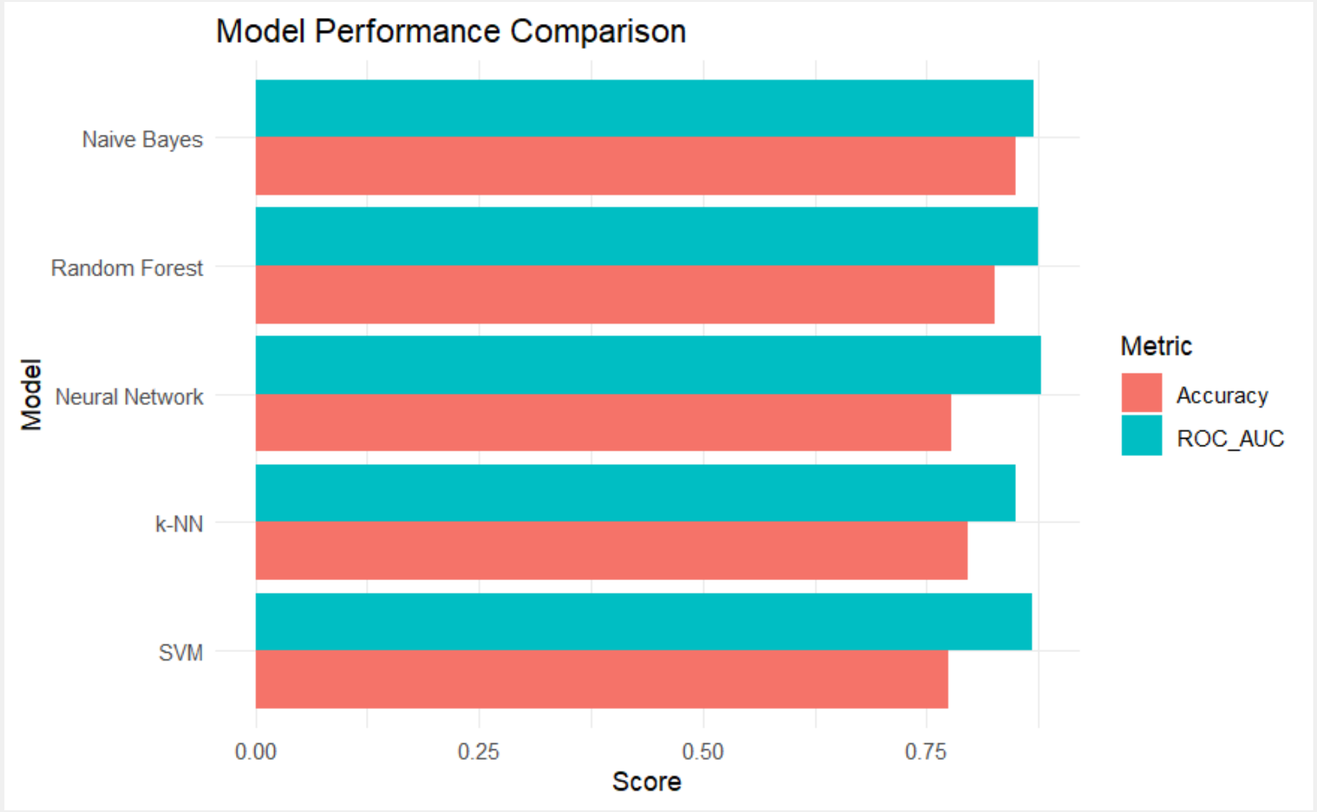
Neural Network

Parâmetro	Valores
hidden_units	16, 32, 64, 128, 256
penalty	0.0001, 0.001, 0.01, 0.1, 1





Comparação de modelos



Conclusão



- O modelo com mais sucesso localmente foi a Neural Network, com um ROC AUC de 0.877
- O pior modelo foi o KNN com um ROC AUC de 0.849
- Seria interessante analisar de que forma se pode incluir a fraud_rate na previsão

initiator <dbl>	fraud_count <int>	total <int>	fraud_rate <dbl>
4.338810e+15	6	13	0.46153846
4.518987e+15	5	17	0.29411765
4.528996e+15	5	17	0.29411765
4.667173e+15	5	13	0.38461538
4.961376e+15	5	17	0.29411765
4.009046e+15	4	18	0.22222222
4.040086e+15	4	14	0.28571429
4.160578e+15	4	14	0.28571429
4.190227e+15	4	14	0.28571429
4.437263e+15	4	8	0.50000000

1-10 of 7,068 rows

Previous 1 2 3 4 5 6 ... 100 Next

Perguntas ?

